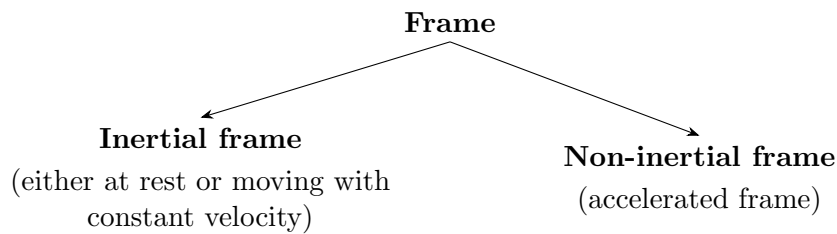


Special Relativity Notes

Frames, Galilean Transformation and Einstein's Postulates

Frame

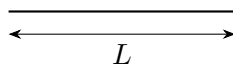
Frame: A frame is a coordinate system.



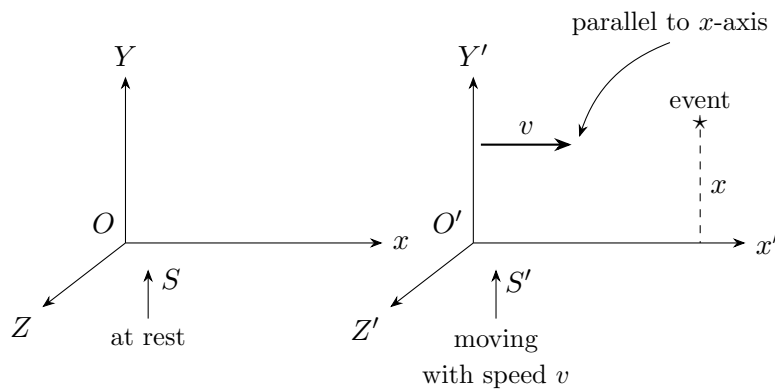
Relativity

Relativity — information shared between observers, one at rest and the other moving with constant velocity (both are in inertial frames).

The fundamental quantities — mass, length and time — remain invariant in our day-to-day life.



Two observers will measure the length of the rod to be the same irrespective of location.



Objective

To describe the coordinates of the event by the two observers S and S' ($v \ll c$).

Galilean coordinate transformation

$$x' = x - vt, \quad y' = y, \quad z' = z, \quad t' = t$$

Galilean velocity transformation

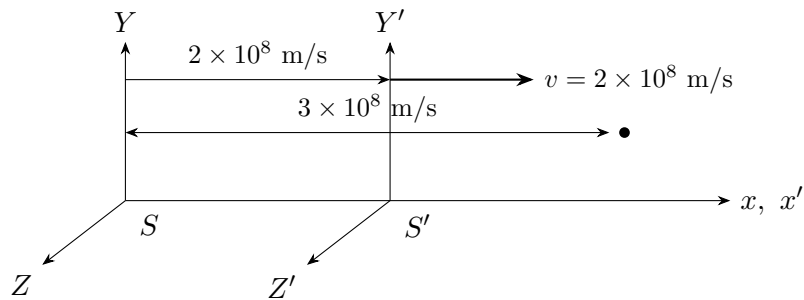
$$v'_x = v_x - v, \quad v'_y = v_y, \quad v'_z = v_z$$

Taking further derivatives:

Galilean transformation for the acceleration and force

$$\begin{aligned} a'_x &= a_x, & a'_y &= a_y, & a'_z &= a_z \\ f'_x &= f_x, & f'_y &= f_y, & f'_z &= f_z \end{aligned}$$

Numerical example



At $t = 0$ both S and S' coincide with each other.

After $t = 1$:

- for S : $3 \times 10^8 \text{ m/s}$
- for S' : $1 \times 10^8 \text{ m/s}$

Einstein's Postulates

- (1) Physics principles have to be the same in all inertial frames of reference.
- (2) The speed of light is the same in all inertial frames of reference.